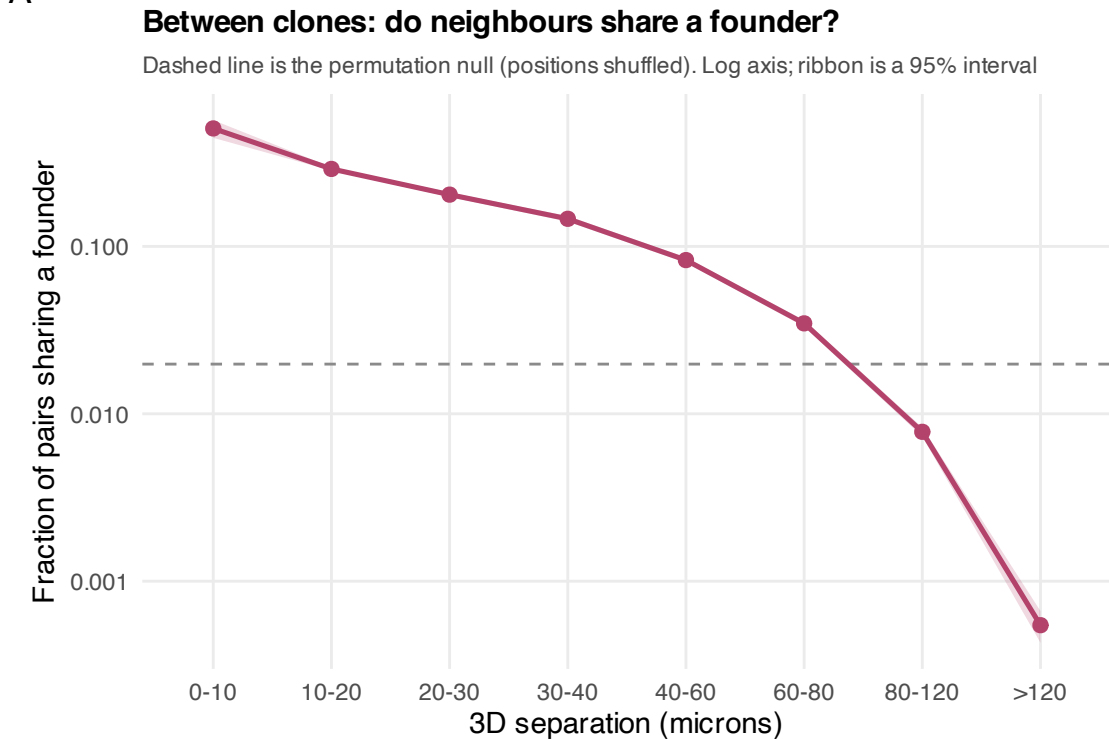


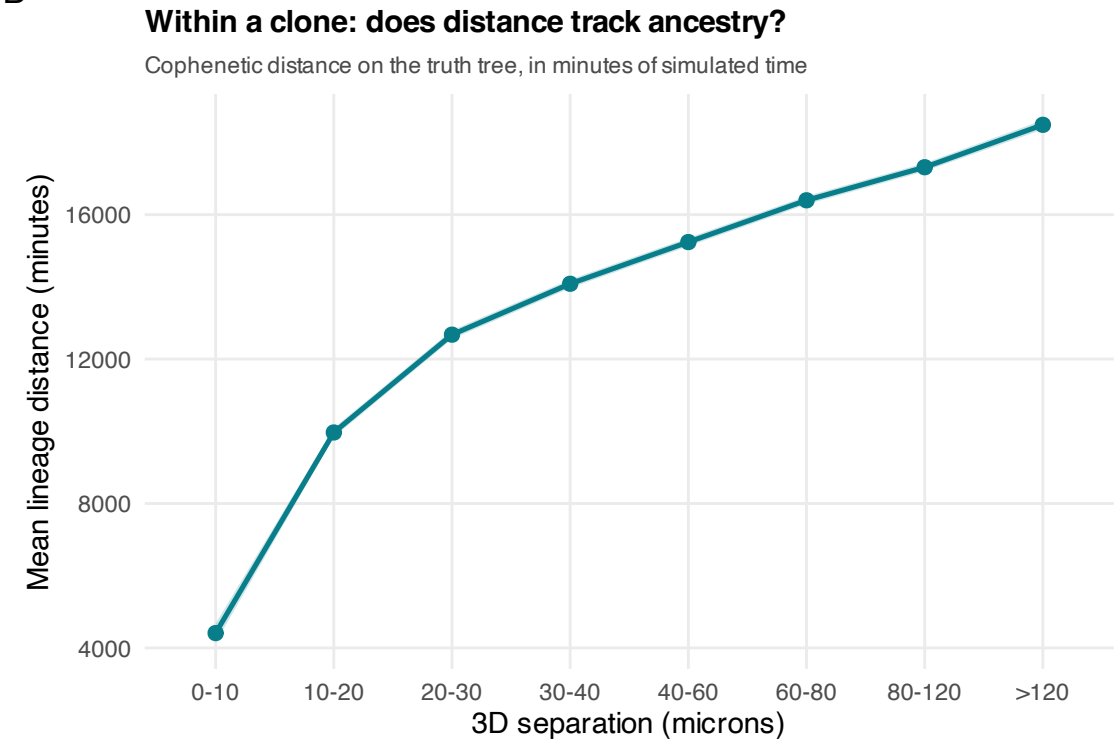
Lineage and spatial organisation in the simulated organoids

Ground-truth lineages and 3D cell positions; no recorder or reconstruction involved

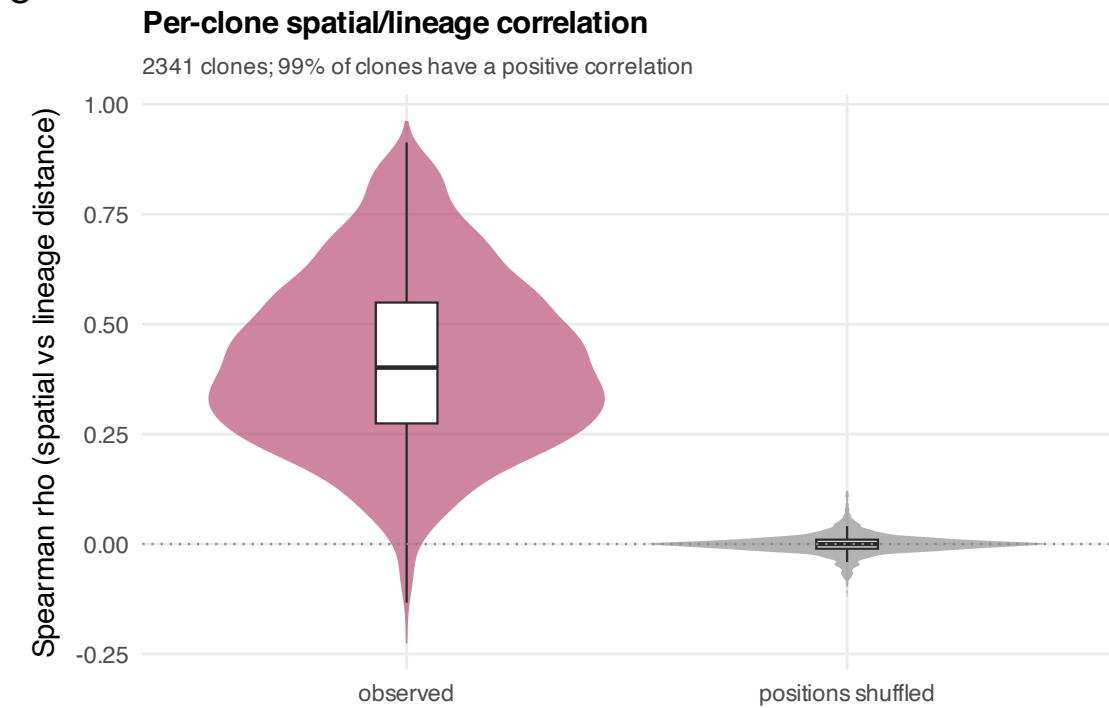
A



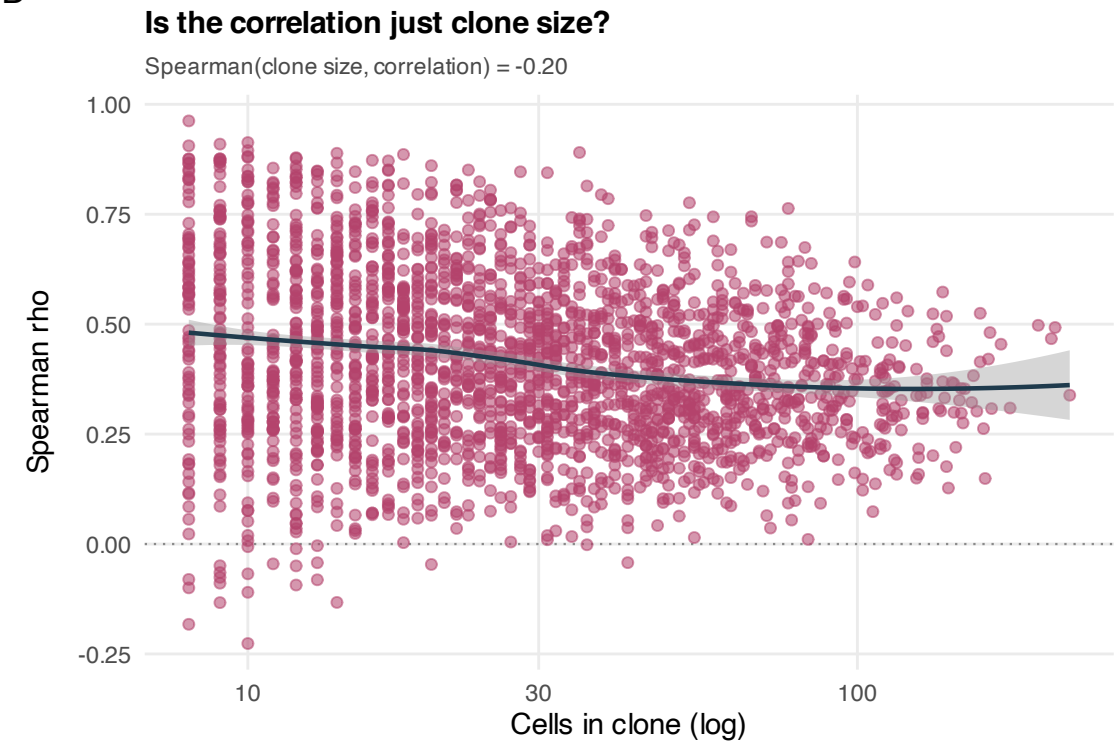
B



C



D



The null shuffles which cell occupies which position, leaving the lineage and the point cloud untouched, so it removes only the correspondence between them. This matters because cells packed in a finite volume are near each other for reasons that have nothing to do with ancestry. Panel A is the clone-territory question and panel B the within-clone question; they are different claims and a recorder could in principle recover one without the other.